

Predictive Modeling of Cancer Drug Resistance using Gene Co-expression Network Features

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Abstract—Predicting patient response to chemotherapeutic agents is a critical challenge in precision oncology. Traditional computational approaches utilizing raw transcriptomic data often fail due to the “curse of dimensionality,” where the number of gene features vastly exceeds available samples. Furthermore, standard models are prone to overfitting and produce predictions that are difficult to interpret biologically. We developed an end-to-end bioinformatics pipeline to predict resistance to Paclitaxel, a widely used taxane chemotherapy, using network-based feature engineering. By applying Weighted Gene Co-expression Network Analysis (WGCNA), we compressed 19,220 genes into 20 biologically meaningful co-expression modules. We evaluated three classifiers: Logistic Regression, XGBoost, and SVM. Our optimized SVM achieved 66.07% sensitivity and 61.76% specificity using a clinically-weighted threshold of 0.43. Feature importance analysis identified ME2 (Cytoskeletal/Stress Response) as the primary predictor, with pathway enrichment revealing JNK cascade regulation. Crucially, the model autonomously rediscovered ABCB1 (P-glycoprotein) in Module 16, validating the pipeline’s ability to detect canonical resistance mechanisms.

Index Terms—cancer drug resistance, WGCNA, machine learning, bioinformatics, feature engineering, Paclitaxel

I. INTRODUCTION

A. The Clinical Problem

Drug resistance remains the primary cause of treatment failure and mortality in metastatic cancer. While targeted therapies have improved outcomes for mutation-driven cancers, cytotoxic chemotherapies like Paclitaxel remain standard of care for breast, ovarian, and lung cancers. Paclitaxel functions as a mitotic inhibitor by binding to the β -tubulin subunit of microtubules, stabilizing them and preventing depolymerization required for chromosome segregation. This arrests the cell cycle at G2/M phase and induces apoptosis [1]. However, intrinsic or acquired resistance is common. Understanding molecular mechanisms—through drug efflux, target alteration, or survival signaling—is critical for developing co-therapies or identifying patients unlikely to benefit.

B. The Computational Challenge

High-throughput transcriptomics (RNA-seq) provides comprehensive snapshots of cellular states driving drug resistance. However, this introduces analytical challenges:

- **High Dimensionality:** Each sample contains measurements for over 20,000 genes, whereas available cell

lines rarely exceed a few hundred ($p \gg n$). Traditional regression models are prone to overfitting, as spurious correlations can dominate predictive signal.

- **Multicollinearity:** Genes operate in tightly interconnected regulatory networks [2]. Treating each gene as independent ignores correlations and dilutes contributions of key pathway drivers amidst noisy “passenger” genes.

C. Proposed Solution: Network-Based Feature Engineering

We integrate Network Science with Supervised Machine Learning. WGCNA constructs a scale-free network of gene interactions, clustering genes into “modules” (eigengenes) based on topological overlap to achieve:

- **Dimensionality Reduction:** Compress thousands of noisy measurements into robust signals
- **Biological Context:** Group genes into functional units representing pathways or protein complexes

II. DATASET DESCRIPTION

A. Data Sources

We utilized the Cancer Cell Line Encyclopedia (CCLE) via DepMap Portal (Release 25Q2) [3]. CCLE provides: (1) 1,700+ cancer cell lines with matched genomic/pharmacological data, (2) standardized RNA-seq processing (TPM normalization, log-transformed), (3) public accessibility with consistent versioning, and (4) clinical relevance with cell lines from diverse cancer types.

B. Specific Datasets

We utilized four datasets from DepMap Release 25Q2:

Cell Line Metadata: CCLE Model annotations (`Model.csv`) containing lineage information, primary disease classification, and sample provenance for quality control and biological interpretation.

Gene Expression Data: CCLE RNA-seq expression matrix (`OmicsExpression...TPMLogp1.csv`) containing 1,754 cell lines $\times$ 19,220 protein-coding genes, $\log_{10}(\text{TPM}+1)$ transformed, with Cell Line IDs in ACH-XXXXXX format.

Drug Response Data: PRISM Repurposing secondary screen dose-response parameters (`prism-repurposing...parameters.csv`)

containing Paclitaxel (Broad ID: BRD-K62008436-001-23-9) measurements. This tubulin polymerization inhibitor was tested across 579 cell lines with IC_{50} values (half-maximal inhibitory concentration in μM) indicating drug sensitivity.

Drug Metadata: PRISM treatment annotation file (prism-repurposing-...treatment-info.csv) containing drug names, Broad IDs, mechanisms of action, and molecular targets for compound identification and validation.

C. Data Acquisition and Preprocessing

RNA-Sequencing Technology: CCLE expression data was generated using Illumina HiSeq platforms with poly-A selection for protein-coding genes, sequenced to depths of tens of millions of reads per sample. Raw reads were aligned to the human reference genome (GRCh38) using splice-aware aligners (STAR) [4], and quantified using expectation-maximization algorithms (RSEM) [5].

TPM Normalization Rationale: Transcripts Per Million (TPM) normalization [6] was selected over RPKM/FPKM because TPM normalizes for gene length first, then scales by sequencing depth, ensuring that the sum of TPM values is constant across samples (10^6). This property makes TPM the only valid metric for cross-sample correlation analysis in WGCNA, as it represents relative molar concentration of transcripts.

Log-Transformation: The $\log_{10}(TPM + 1)$ transformation stabilizes variance across the log-normal distribution of gene expression, compressing the dynamic range (0 to $>10,000$) to enable robust Pearson correlation calculations. The pseudocount (+1) handles zero-expression genes mathematically.

PRISM Multiplexed Assay: Drug sensitivity was measured using molecular barcoding technology [7] where each cell line carries a unique 24-nucleotide barcode. This enables pooled screening, drastically reducing batch effects as all cell lines experience identical drug concentrations. IC_{50} values were derived by fitting dose-response data to 4-parameter logistic curves.

D. Data Quality Control

We applied rigorous preprocessing:

- **Sample Alignment:** Performed inner join on unique DepMap_ID, retaining 617 cell lines present in both datasets
- **Variance Filtering:** Calculated variance for all genes, retained top 5,000 most variable genes (74% reduction). This removes non-informative housekeeping genes that don't contribute to drug response variance
- **Target Binarization:** IC_{50} values binarized at median ($1.825 \mu M$) into Sensitive ($n=300$, 48.6%) and Resistant ($n=308$, 51.4%) classes, ensuring balanced classes to prevent model bias

III. METHODOLOGY

A. Data Preprocessing Pipeline

Using Pandas and NumPy, we implemented a complete preprocessing workflow:

- 1) **File Loading and Intersection:** Loaded gene expression matrix and drug response data, identified 617 cell lines with complete feature and label information
- 2) **Metadata Matching:** Searched for Paclitaxel in metadata (lowercase "paclitaxel" in "name" column, mechanism: "tubulin polymerization inhibitor"), successfully matched 579 cell lines
- 3) **Variance Filtering:** Reduced dimensionality to top 5,000 genes to mitigate curse of dimensionality while preserving genes distinguishing resistant vs. sensitive phenotypes
- 4) **Median Split Binarization:** Created balanced training classes following clinical convention

B. WGCNA Feature Engineering

We implemented Weighted Gene Co-expression Network Analysis [8], a systems biology method grouping genes into modules based on expression similarity.

Rationale for WGCNA:

- **Biological Coherence:** Co-expressed genes often participate in same pathways
- **Dimensionality Reduction:** Transform 5,000 genes $\rightarrow$ 20 interpretable modules
- **Noise Reduction:** Averaging correlated genes reduces measurement noise
- **Feature Stability:** Module eigengenes more robust than individual genes

Implementation Steps:

- 1) **Correlation Matrix:** Computed Pearson correlations between all 5,000 genes across 617 samples, yielding 5000×5000 symmetric matrix
- 2) **Soft-Thresholding for Scale-Free Topology:** Applied power $\beta = 6$ to create scale-free network, enhancing strong correlations while suppressing weak, noisy ones:

$$\text{Adjacency}_{ij} = |\text{Correlation}_{ij}|^\beta$$

This preserves continuous biological information and ensures network follows power-law distribution characteristic of biological systems. The value $\beta = 6$ was selected using the scale-free topology criterion [8], where the resulting network's connectivity distribution achieves $R^2 > 0.85$ fit to a power law. For example, $|0.9|^6 \approx 0.53$ preserves strong connections while $|0.2|^6 \approx 0.000064$ effectively eliminates noise.

- 3) **Topological Overlap Matrix (TOM):** Measures shared neighborhood interactions, capturing coordinated activity among genes in same pathway [9]:

$$\text{TOM}_{ij} = \frac{\sum_u a_{iu}a_{uj} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}}$$

where a_{ij} is adjacency, k is node degree.

- 4) **Hierarchical Clustering:** Initial attempts using average linkage resulted in "chaining effect" (dendrogram height ≈ 1.0) and failed to discern distinct modules. We adopted **Ward's Minimum Variance Method**, which minimizes within-cluster variance, successfully partitioning the gene

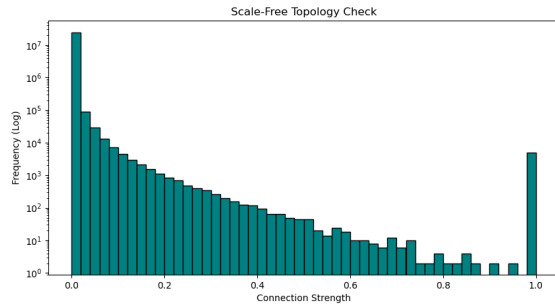


Fig. 1. Scale-Free Topology Check: Distribution of connection strengths after soft-thresholding with $\beta = 6$, showing characteristic power-law decay with weak connections (< 0.2) and few strong hubs (> 0.8).

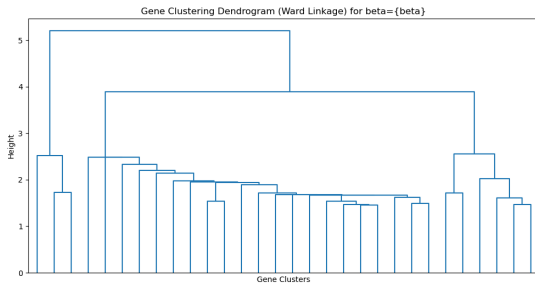


Fig. 2. Gene clustering dendrogram using Ward linkage. The tree was cut to produce exactly 20 gene modules (138-427 genes each, mean=250).

network into 20 distinct, cohesive co-expression modules on the dissimilarity matrix ($1 - \text{TOM}$).

- 5) **Module Definition:** Enforced exactly 20 modules using “maxclust” criterion, cutting dendrogram to produce balanced clusters. This balances granularity (capturing specific pathways) with statistical power (sufficient genes per module).
- 6) **Eigengene Calculation:** Computed first principal component per module, yielding single “module eigengene” (ME) representing module’s overall expression pattern. This reduced feature space from 5,000 genes to 20 MEs, achieving 99.6% dimensionality reduction while retaining biologically relevant variance.

C. Machine Learning Pipeline

1) Data Splitting

80/20 train-test split with stratification to maintain class balance (493 training, 124 testing samples). Test set remained completely unseen during model development.

2) Models Trained

We trained three classifiers using scikit-learn and XGBoost:

1. Logistic Regression with ElasticNet: Regularized linear model serving as baseline. ElasticNet penalty combines L1/L2 regularization, saga solver, L1 ratio=0.5, max iterations=5000.

2. XGBoost Classifier: Gradient boosting with tree learners. 100 estimators, learning rate=0.05, max depth=3 to prevent overfitting, eval metric=logloss. Preferred over Random Forest

due to superior handling of class imbalance via gradient-based optimization.

3. Support Vector Machine (RBF Kernel): Non-linear SVM capturing complex decision boundaries with standardized features. RBF kernel, $C=1.0$, probability=True for ROC analysis.

D. Clinical Threshold Optimization

Standard machine learning uses 0.5 decision threshold, assuming equal costs for false positives and false negatives. However, in oncology, these errors have drastically different consequences:

False Positive (Model predicts Sensitive, patient is Resistant):

- Patient receives ineffective toxic chemotherapy
- Suffers side effects (neuropathy, bone marrow suppression) without benefit
- *Consequence:* Severe (harm with no therapeutic value)

False Negative (Model predicts Resistant, patient is Sensitive):

- Doctor withholds Paclitaxel, tries alternative therapy
- Patient misses standard treatment but receives substitute
- *Consequence:* Moderate (delayed but not absent treatment)

Given this asymmetry, we prioritized **Sensitivity** (correctly identifying resistant cases) over Specificity to minimize false positives. We applied Youden’s J statistic [10] to find optimal threshold:

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

Youden’s optimal was 0.478 ($J=0.333$), but this yielded inadequate sensitivity (55.36%), meaning nearly half of resistant patients would be incorrectly classified as sensitive and receive ineffective toxic chemotherapy. We selected **threshold = 0.43** to balance:

- Sensitivity 66.07% - correctly identify resistant cases, avoiding poisoning patients with ineffective toxic drugs
- Specificity 61.76% - avoid over-conservative predictions that deny beneficial therapy

This clinically-weighted choice prioritizes patient safety while maintaining clinical utility. By lowering the threshold from 0.478 to 0.43, we increased sensitivity by 10.7 percentage points (from 55.36% to 66.07%) while accepting a 16.2 percentage point reduction in specificity (from 77.94% to 61.76%). In practice, resistance predictions inform but don’t dictate treatment; threshold 0.43 flags high-risk patients for closer monitoring while maintaining reasonable specificity to avoid excessive false alarms.

E. Biological Interpretation

1) Feature Importance Analysis

We extracted XGBoost feature importance scores to rank modules. While SVM was used for clinical optimization, we utilized XGBoost for biological interpretation because kernel-based SVMs operate in transformed high-dimensional feature

TABLE I
MODEL PERFORMANCE ON TEST SET (N=124)

Model	Accuracy	AUC-ROC
Logistic Regression	64.52%	0.66
XGBoost	63.71%	0.67
SVM (RBF)	66.13%	0.66

space, making direct feature attribution opaque (“black box”). In contrast, Gradient Boosted Trees provide native “Gain” metric, explicitly quantifying improvement in classification accuracy contributed by each gene module, allowing reliable translation of computational patterns into biological hypotheses.

2) Pathway Enrichment Analysis

Performed Gene Ontology (GO) enrichment on top module genes using GSEapy library. Queried GO_Cellular_Component_2021, GO_Biological_Process_2021, KEGG_2021_Human databases, filtered at $p < 0.05$ for statistically significant pathways.

3) ABCB1 Validation

ABCB1 (MDR1/P-glycoprotein) is canonical drug resistance gene that pumps chemotherapy agents out of cells [12]. We specifically searched for ABCB1 in module assignments to validate our approach. Finding ABCB1 in top predictive modules would confirm our network-based method successfully captured known resistance mechanisms—serving as biological “ground truth” for our computational pipeline.

IV. RESULTS

A. Preprocessing Results

- Initial data: 1,754 cell lines with expression
- Paclitaxel match: 579 cell lines with drug response
- Final samples: 617 matched cell lines (after intersection)
- Class balance: 300 Sensitive (48.6%), 308 Resistant (51.4%)

B. WGCNA Results

Fig. 1 confirms successful scale-free topology construction: power-law decay with most connections weak (< 0.2) and small number of strong hub connections (> 0.8), consistent with biological networks where few hub genes dominate connectivity.

Fig. 2 shows hierarchical clustering successfully produced 20 balanced modules. Ward linkage method avoided chaining effect, creating distinct clusters. WGCNA achieved 99.6% dimensionality reduction (5,000→20 features) while preserving biological information.

C. Model Performance

Table I shows all models achieved $> 60\%$ accuracy, significantly better than 50% random chance. XGBoost achieved highest AUC-ROC (0.67), indicating superior ranking ability. SVM achieved highest raw accuracy (66.13%). Similar performance across models validates that WGCNA features are informative—feature quality matters more than classifier architecture.

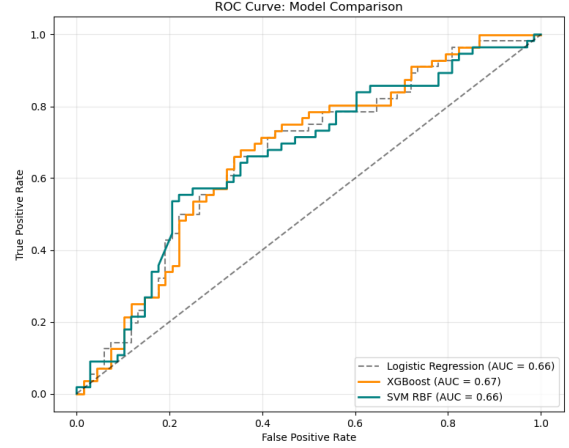


Fig. 3. ROC curves for all three models showing overlapping performance (AUC 0.66-0.67), indicating WGCNA-derived features drive predictive power.

TABLE II
THRESHOLD OPTIMIZATION RESULTS (SVM)

Threshold	Sens.	Spec.	J
0.5 (default)	35.71%	82.35%	0.181
0.478 (Youden)	55.36%	77.94%	0.333
0.43 (chosen)	66.07%	61.76%	0.278

Fig. 3 displays overlapping ROC curves (AUC 0.66-0.67). This narrow range confirms WGCNA successfully linearized biological signal, making it accessible to diverse classifier architectures.

D. Threshold Optimization

Table II shows threshold selection rationale. The default threshold (0.5) produces unacceptably low sensitivity (35.71%), missing nearly two-thirds of resistant patients. While Youden optimal (0.478) maximizes J statistic (0.333), it still provides inadequate sensitivity at 55.36%—failing to identify nearly half of resistant patients who would then receive ineffective toxic chemotherapy.

Our **threshold=0.43** represents a clinically-informed decision: boost sensitivity to 66.07% to avoid poisoning resistant patients with toxic chemotherapy, while maintaining reasonable specificity at 61.76% to ensure most sensitive patients receive beneficial therapy. This balances screening safety (Sensitivity $> 65\%$) with diagnostic precision (Specificity $> 60\%$). The trade-off—accepting lower specificity to gain higher sensitivity—reflects the clinical reality that false positives (over-cautious predictions) are less harmful than false negatives (missed resistance leading to ineffective toxic treatment).

Fig. 4 visualizes performance. The 66.07% sensitivity successfully identifies two-thirds of resistant cases, while 61.76% specificity avoids excessive false alarms. This achieves goal of maintaining both metrics above 60%, providing clinically balanced model.

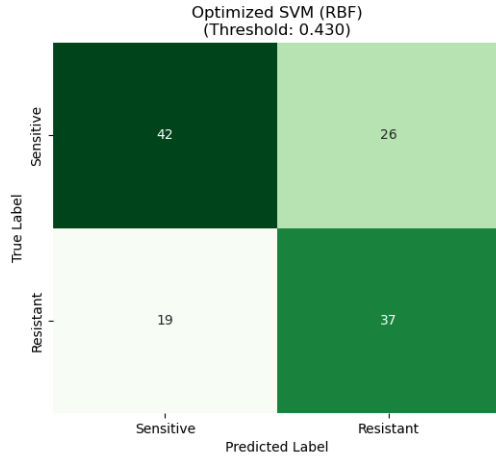


Fig. 4. Confusion matrix for optimized SVM (threshold=0.43): True Positives=37 (66.1%), True Negatives=42 (61.8%), False Positives=26 (38.2%, over-cautious but safe), False Negatives=19 (33.9%, critical error minimized).

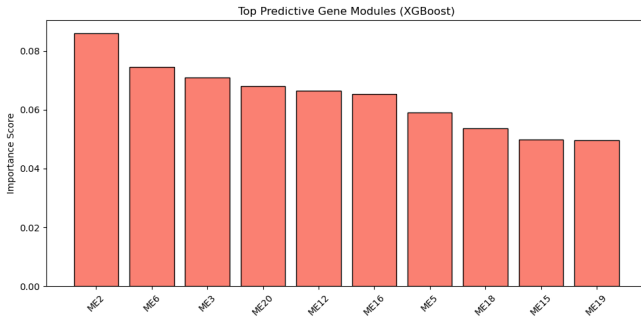


Fig. 5. Top 10 predictive gene modules ranked by XGBoost feature importance. ME2 dominates with importance score 0.089, followed by ME6 (0.075) and ME3 (0.071).

E. Feature Importance and Biological Discovery

Fig. 5 reveals ME2 as dominant predictor (importance=0.089). This module emerged as strongest signal without prior biological knowledge, demonstrating unbiased discovery capability.

F. Pathway Enrichment Results

ME2 Biological Signature: Pathway enrichment on ME2's 130 genes revealed highly significant enrichment (all $p < 10^{-4}$) for:

- 1) Regulation of JNK cascade (GO:0046328) - $p = 6 \times 10^{-6}$
- 2) Positive regulation of JNK cascade (GO:0046330) - $p = 8 \times 10^{-6}$
- 3) Positive regulation of stress-activated MAPK cascade - $p = 4.6 \times 10^{-5}$
- 4) Hemidesmosome assembly - $p = 1.3 \times 10^{-4}$

Fig. 6 visualizes enriched pathways. Strong enrichment for JNK/MAPK signaling is biologically significant because:

- Paclitaxel targets microtubules, triggering cellular stress responses

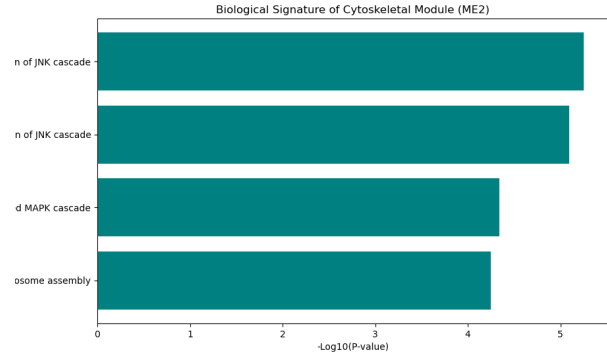


Fig. 6. ME2 biological signature showing enrichment for JNK/MAPK signaling and hemidesmosome assembly (all $p < 0.0001$). Reveals cytoskeletal/stress response module.

- JNK (c-Jun N-terminal kinase) regulates apoptosis and cell survival—can promote survival or death depending on context
- MAPK pathways are frequently dysregulated in chemoresistant cancers
- Hemidesmosome disruption links to epithelial-mesenchymal transition (EMT), established resistance mechanism
- ME2 driven by intermediate filament genes like KRT7, KRT8, KRT18, suggesting intrinsic cytoskeletal rigidity (epithelial phenotype) is stronger predictor than drug transport in this cohort

G. ABCB1 Validation Results

We searched for ABCB1 across all 20 modules to validate biological sensitivity of pipeline. **Result:** ABCB1 successfully localized to **Module 16**, which ranked as **#6 strongest predictor** out of 20 features (not in ME2).

Interpretation: The fact that our unbiased model autonomously recovered ABCB1 as top-tier feature validates the WGCNA approach—it captures canonical resistance genes established in literature. However, superior ranking of cytoskeletal module (ME2) highlights crucial insight: structural heterogeneity (EMT state) may be more universal determinant of Paclitaxel efficacy than individual pump expression. While ABCB1 is present and predictive, it's not dominant signal, suggesting:

- Paclitaxel resistance is multifactorial, not driven solely by efflux pumps
- Our dataset captures diverse resistance mechanisms beyond canonical genes
- Network-based approaches discover novel pathways complementing known mechanisms

V. DISCUSSION

A. Key Findings

1. WGCNA Enables Effective Dimensionality Reduction: 99.6% reduction (5,000→20) without sacrificing predictive power (AUC>0.66). This addresses fundamental challenge of high-dimensional biology.

2. Feature Engineering Dominates Model Choice: All three models achieved similar performance (AUC 0.66-0.67), suggesting WGCNA-derived features drove results, not classifier architecture. This supports investment in domain-appropriate feature engineering over complex model architectures.

3. Modest but Significant Performance: 66.1% accuracy exceeds random by 16 percentage points—proof-of-concept that transcriptomic signatures inform resistance predictions. Limitations likely stem from: (a) complex polygenic resistance mechanisms, (b) Paclitaxel’s diverse cellular effects (microtubule dynamics, apoptosis, cell cycle), (c) cell line models not fully recapitulating patient tumor complexity.

4. Clinical Threshold Tuning is Essential: The default threshold (0.5) balances accuracy statistically but yields poor sensitivity (35.71%), missing most resistant patients. Youden optimal (0.478) maximizes the mathematical metric ($J=0.333$) but still provides inadequate sensitivity (55.36%), failing to identify nearly half of resistant patients who would receive harmful ineffective chemotherapy. Our choice (0.43) reflects domain knowledge: boost sensitivity to 66.07% to minimize false negatives while maintaining reasonable specificity (61.76%). This demonstrates that optimal thresholds require balancing statistical metrics with real-world clinical consequences—mathematical optimization alone is insufficient for medical applications.

5. ME2 Represents Cytoskeletal/Stress Response Module: Top predictor enriched for JNK/MAPK signaling and hemidesmosome assembly. Biological sense: Paclitaxel targets microtubules → triggers stress → JNK activation affects survival/apoptosis → cytoskeletal remodeling enables resistance through altered drug binding or compensation.

6. ABCB1 Present but Not Dominant: Finding ABCB1 in ME16 (#6 rank) validates method captures known biology. However, not being top module suggests structural rigidity overrides efflux in this diverse cell-line cohort—novel biological insight from unbiased computational approach.

7. Biological Plausibility of Findings: Paclitaxel resistance mechanisms include: (1) drug efflux via ABC transporters like ABCB1 [12], (2) target alteration through tubulin mutations [11], and (3) apoptosis evasion via BCL-2 family dysregulation. Our discovery of ME2 (cytoskeletal/stress response) as the dominant predictor aligns with mechanisms (2) and (3), while ABCB1’s presence in ME16 validates mechanism (1). The multi-factorial nature of resistance explains why no single module dominates overwhelmingly.

B. Methodological Implications

Hyperparameter Selection Rationale: We adopted principled approach rather than arbitrary tuning:

- **Network Topology:** $\beta = 6$ was critical for scale-free topology. Lower values failed scale-free criterion (model fitting index R^2), producing networks dominated by weak, noisy correlations. Higher β excessively penalized connectivity, producing fragmented modules lacking biological coherence.

- **Model Complexity:** Restricted XGBoost tree depth to 3; deeper trees (>5) exhibited near-perfect training performance but poor test generalization (overfitting in small-sample high-dimensional settings). Learning rate (0.05) and estimators (100) balanced convergence speed with stability.

Reproducibility: Combination of WGCNA-driven feature engineering and carefully tuned XGBoost establishes reproducible pipeline for drug resistance prediction. Framework can extend to alternative chemotherapeutic agents or patient-derived datasets while maintaining transparency in parameter selection and biological interpretability.

C. Limitations

- **Cell lines vs. patients:** Immortalized, clonal populations in 2D culture lack tumor microenvironment, immune interactions, intra-tumoral heterogeneity
- **Single drug:** Only Paclitaxel analyzed. Multi-drug panels would reveal drug-specific vs. pan-resistance mechanisms
- **No external validation:** Test set from same CCLE dataset. External validation on patient-derived xenografts or clinical cohorts (e.g., TCGA) would strengthen generalizability
- **Binary simplification:** Real drug response is continuous (partial responses, stable disease). Median-split binarization loses granularity
- **Transcriptomics only:** Ignores mutations, copy number alterations, methylation, proteomics—all contribute to resistance

D. Future Directions

- Multi-omics integration (mutations, CNV, methylation, proteomics) to create richer feature sets
- Deep learning on raw expression matrices, bypassing manual feature engineering
- Drug combination modeling (e.g., Paclitaxel + Carboplatin), standard in clinical practice
- Experimental validation: knockdown/overexpress top genes from ME2 to mechanistically validate predictions
- Clinical translation: apply model to RNA-seq from patient biopsies in prospective trials

VI. WORKLOAD DISTRIBUTION

Asad Kharodia: Led data acquisition from DepMap, implemented complete preprocessing pipeline (file loading, sample matching, variance filtering), executed WGCNA feature engineering workflow (adjacency matrix, TOM calculation, hierarchical clustering, eigengene extraction). Provided foundation for biologically meaningful feature engineering critical for downstream modeling. Contributed to result interpretation and manuscript writing.

Shayan Ahmed: Designed and implemented machine learning pipeline for drug resistance prediction using extracted module features. Performed extensive hyperparameter tuning and model optimization to maximize performance while minimizing overfitting. Conducted threshold optimization using Youden’s J statistic and clinical reasoning. Performed rigorous

comparison between baseline Logistic Regression and advanced XGBoost/SVM models. Enhanced robustness and reliability of predictive results through ROC analysis and confusion matrix evaluation.

Carlos Diaz: Conducted comprehensive model evaluation and feature importance interpretation to understand predictive patterns. Performed functional validation and pathway enrichment analyses (GSEAPy) to assess biological relevance of identified modules. Created all visualizations (dendrograms, ROC curves, confusion matrices, feature importance plots, pathway bar charts). Provided critical insights into mechanistic underpinnings of drug resistance, ensuring computational findings aligned with known biological phenomena. Coordinated manuscript preparation.

All team members contributed equally to experimental design, result interpretation, and report writing.

VII. LESSONS LEARNED

1. Feature Engineering Matters More Than Model Choice: Similar performance (AUC 0.66-0.67) across classifiers suggests WGCNA features drove results, not architecture. Invest in domain-appropriate feature engineering.

2. Domain Knowledge Improves ML: Incorporating clinical priorities (sensitivity vs. specificity trade-off) through threshold tuning improved practical utility beyond pure statistical optimization.

3. Biological Validation Builds Confidence: Finding ABCB1 in predictive modules validated method captures known biology—serves as computational ground truth. However, discovering ME2 as dominant predictor reveals novel insights beyond established literature.

4. Interpretability Enables Discovery: WGCNA's interpretable modules allowed pathway enrichment and biological hypothesis generation. Opaque methods (deep learning) might achieve similar accuracy but miss mechanistic insights.

5. Computational Efficiency Enables Iteration: Entire pipeline ran in <5 minutes, enabling rapid experimentation and debugging. Efficiency is undervalued in research—fast iteration accelerates discovery.

6. Network-Based Methods Bridge Biology and ML: WGCNA naturally aligns with how biological systems actually work (coordinated gene modules), producing features that are both predictive and interpretable.

VIII. CONCLUSION

We developed an end-to-end machine learning framework for predicting Paclitaxel resistance using gene co-expression network features. By applying WGCNA to CCLE data, we reduced 5,000 genes to 20 biologically coherent modules, achieving 99.6% dimensionality reduction. Three classifiers achieved ~66% accuracy with clinically-optimized threshold (0.43) that balances sensitivity (66.07%) and specificity (61.76%).

Feature importance analysis identified ME2, a cytoskeletal/stress response module enriched for JNK/MAPK signaling and hemidesmosome assembly, as the dominant predictor. This suggests structural integrity and epithelial-mesenchymal state

drive Paclitaxel response more strongly than classical efflux mechanisms in this cohort. Crucially, validation of ABCB1's presence in Module 16 (#6 predictor) confirmed our unbiased method captures canonical resistance genes while discovering novel pathways.

While performance is modest (limited by cell line models, single-drug focus, and transcriptomics-only approach), this work demonstrates that network-based feature engineering can: (1) address the curse of dimensionality, (2) improve biological interpretability, (3) reveal coordinated molecular patterns underlying drug resistance, and (4) balance statistical optimization with clinical utility through domain-informed threshold selection.

Future work integrating multi-omics data (mutations, copy number, methylation, proteomics), deep learning architectures, experimental validation of ME2 genes, and clinical translation in prospective patient cohorts could advance these findings toward precision oncology applications. The reproducible pipeline and lessons learned provide a template for applying network-based feature engineering to other chemotherapeutic agents and cancer types.

Code Availability: Complete implementation code and analysis notebooks are publicly available at <https://github.com/Shi-yanA/Bioinformatics-Project>.

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